

System Administration and Maintenance

Project Title:
**Enterprise Server Deployment and Operating
System Installation**

Submitted by:

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Project Overview

Operating systems form the core foundation of enterprise IT infrastructure, enabling servers to host web services, databases, and business applications. In this project, acting in the role of a Junior System Administrator, an Ubuntu Server instance was deployed and configured for a startup environment. This documentation covers the end-to-end deployment process, including operating system installation, post-install configuration, system verification, and a technical comparison of modern boot architectures.

Objectives

- **Deploy a Production-Ready OS:** Perform a complete, headless installation of Ubuntu Server 24.04 LTS within a VirtualBox virtualized environment.
- **Establish Network & Remote Access:** Configure essential network interfaces, verify internet reachability, and ensure the OpenSSH service is active for secure remote administration.
- **Validate System Health:** Execute command-line verification tools to confirm hostname identity, IP configuration, and account privileges (sudo).
- **Ensure System Maintenance:** Refresh package repository indexes and apply the latest system updates (apt update / apt upgrade).
- **Technical Analysis & Documentation:** Evaluate boot architectures (BIOS vs. UEFI), compare server operating systems, and document the overall deployment process for enterprise knowledge sharing.

Software Requirements

- Host Operating System: Windows 10/11, macOS, or Linux (64-bit)
- Hypervisor / Virtualization Software: Oracle VM VirtualBox (v7.0 or higher)
- Guest Operating System ISO: Ubuntu Server 24.04 LTS (ubuntu-24.04-live-server-amd64.iso)
- SSH Terminal Client: Windows PowerShell, Command Prompt, or OpenSSH Client (for remote terminal management)

- Web Browser: Google Chrome / Microsoft Edge (for downloading installer media and accessing GitHub)

Virtual Machine Specifications

- VM Name: Ubuntu-Server-Week03
- Guest OS Type: Ubuntu (64-bit)
- Processor (vCPU): 2 Cores
- Base Memory (RAM): 4096 MB (4 GB)
- Controller Type: SATA Controller
- Storage: 40 GB
- Network Adapter Mode: NAT (Network Address Translation)
- Boot Order: Floppy, Optical, Hard Disk
- Graphics / Video Memory: 16 MB

Step-by-Step Installation Procedure

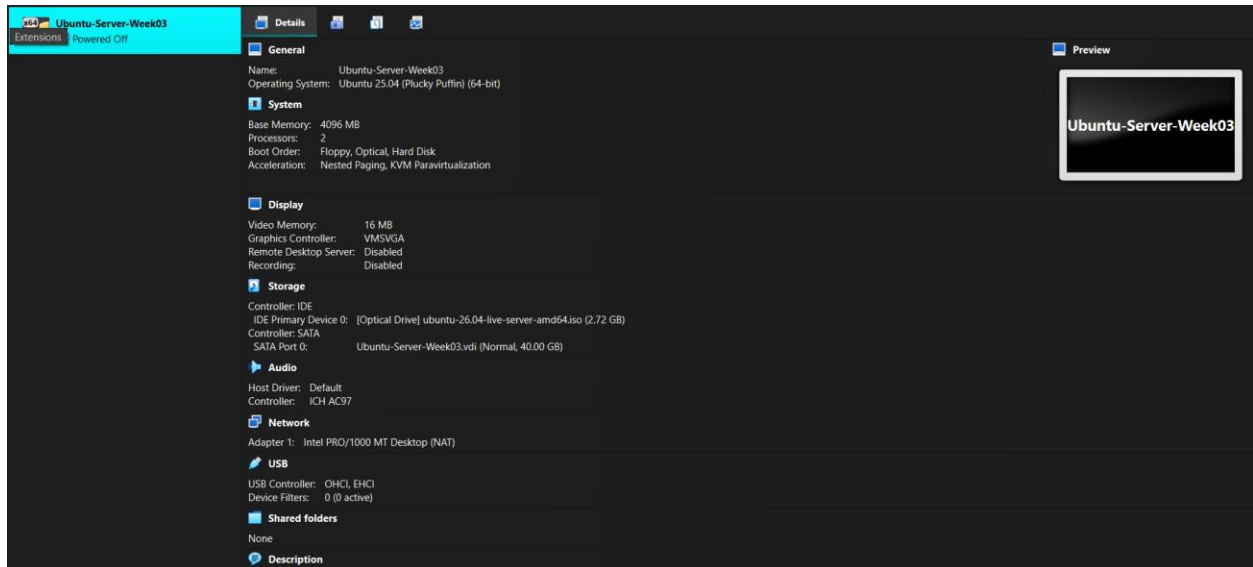
UBUNTU SERVER INSTALLATION

1. Create the Virtual Machine

Open Oracle VM VirtualBox and create a new virtual machine with the following configuration:

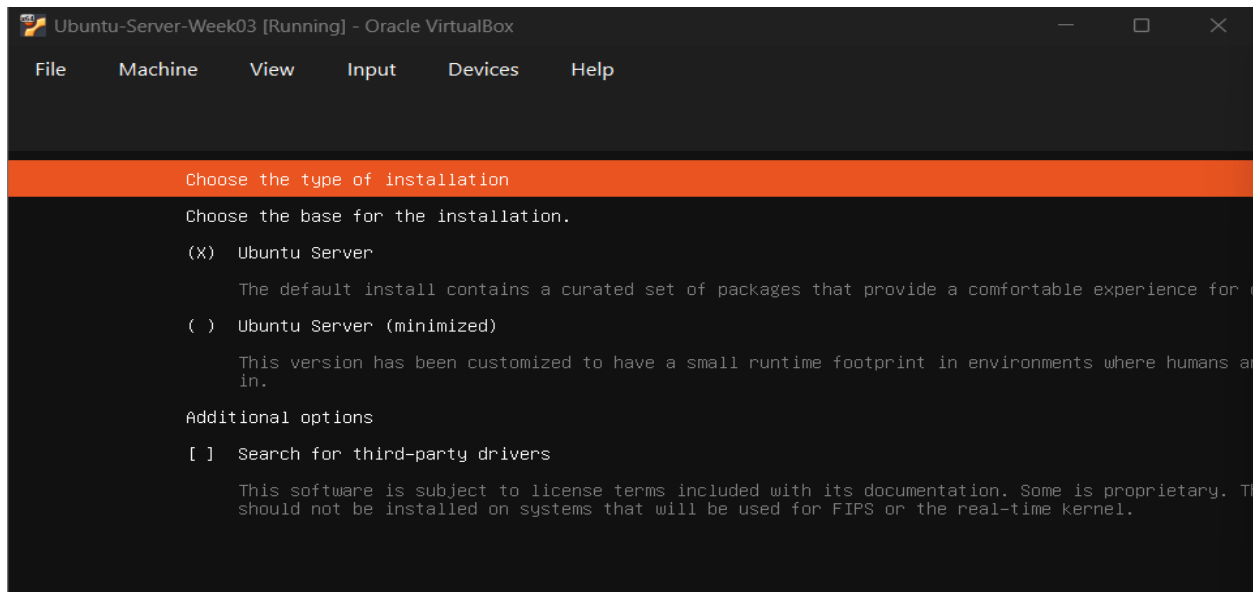
- VM Name: Ubuntu-Server-Week03
- ISO Image: Ubuntu Server ISO
- Installation Type: Manual installation / unattended installation disabled
- Memory: 4096 MB (4 GB)
- Processors: 2
- Virtual Hard Disk: 40 GB

- Network: NAT



2. Start the Virtual Machine

- Start the Ubuntu-Server-Week03
- Select “Ubuntu Server” when prompted for Installation type



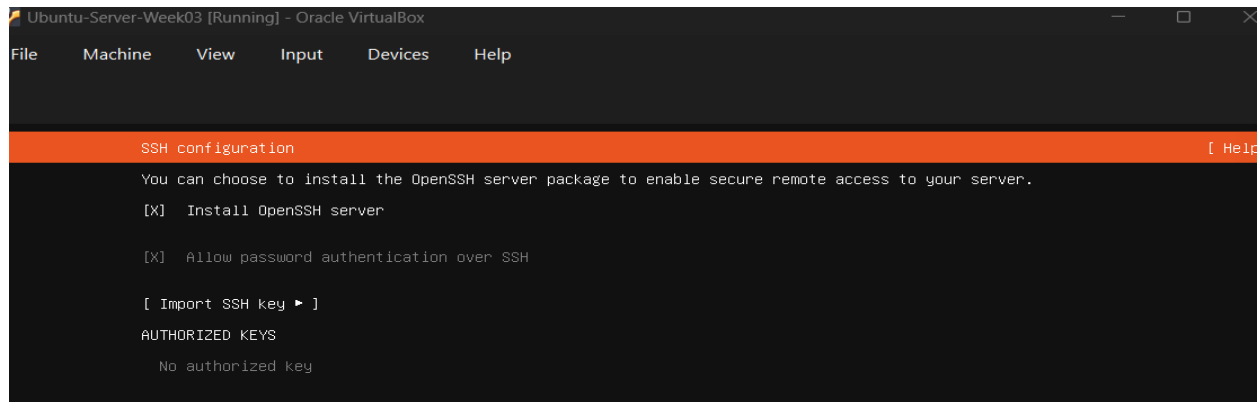
3. Profile Configuration

- Set the administrator's name. Example: “YourName”
- Set the host name. Example: “server01”

- Set username. Example: “lastname”
- Set the password.

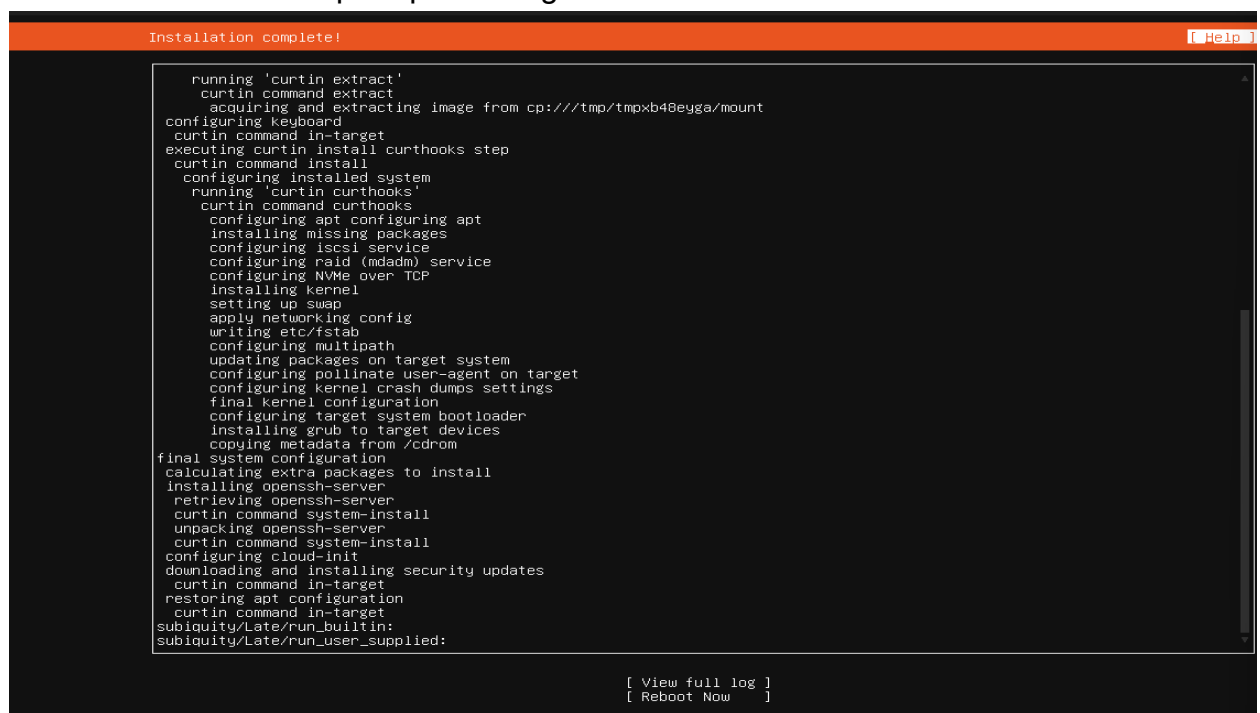
4. SSH Configuration

- Make sure the “Install OpenSSH server” selected



5. After the installation

- Make sure the installation is completed.
- After completion, press “Reboot Now”.
- The VM will shut down/restarts.
- You will then be prompted to login.



6. Enter credentials

- Enter all your login credentials to proceed.
- After login, enter “hostname”.

```
Ubuntu-Server-Week03 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Ubuntu 26.04 LTS server01 tty1
server01 login: quizon
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-29-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Aug 14 02:50:24 PM UTC 2026

System load:          1.22
Usage of /:            20.0% of 39.07GB
Memory usage:         6%
Swap usage:           0%
Processes:            118
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:feaa:f72d

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

quizon@server01:~$
```

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

quizon@server01:~$ hostname
server01
quizon@server01:~$
```

7. Verify the IP

- Type “ip addr” to see the assigned IP address.

```
quizon@server01:~$ hostname
server01
quizon@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:aa:f7:2d brd ff:ff:ff:ff:ff:ff
    altname enx080027aaf72d
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85958sec preferred_lft 85958sec
    inet6 fd17:625c:f037:2:a00:27ff:feaa:f72d/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86182sec preferred_lft 14182sec
    inet6 fe80::a00:27ff:feaa:f72d/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
quizon@server01:~$
```

8. Test the Internet Connectivity

- Type “ping -c 4 google.com”
- You should receive successful replies or no packet loss.

```
quizon@server01:~$ ping -c 4 google.com
PING google.com (172.217.26.78) 56(84) bytes of data:
64 bytes from sin10s02-in-f14.1e100.net (172.217.26.78): icmp_seq=1 ttl=64 time=11.3 ms
64 bytes from sin10s02-in-f14.1e100.net (172.217.26.78): icmp_seq=2 ttl=64 time=16.0 ms
64 bytes from sin10s02-in-f14.1e100.net (172.217.26.78): icmp_seq=3 ttl=64 time=16.3 ms
64 bytes from sin10s02-in-f14.1e100.net (172.217.26.78): icmp_seq=4 ttl=64 time=16.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 6676ms
rtt min/avg/max/mdev = 11.304/15.082/16.721/2.197 ms
quizon@server01:~$
```

9. Update the Server

- Type “sudo apt update” and “sudo apt upgrade -y”
- You can also do “sudo apt update && sudo apt upgrade -y”
- The update should start.

```
Ubuntu-Server-Week03 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

get:41 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [640 B]
get:42 http://archive.ubuntu.com/ubuntu resolute-backports/universe Translation-en [300 B]
get:43 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [216 B]
get:44 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 c-n-f Metadata [116 B]
get:45 http://archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 Components [216 B]
get:46 http://archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 c-n-f Metadata [120 B]
Fetched 32.3 MB in 11s (2,843 kB/s)
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14          linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14          pollinate
Use 'sudo apt autoremove' to remove them.

Upgrading:
  linux-firmware-amd-misc

Not upgrading yet due to phasing:
  python3-software-properties  software-properties-common

Summary:
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 2
  Download size: 584 kB
  Freed space: 1,024 B

get:1 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-amd-misc all 20260319.git217ca6e4-0ubuntu1.1 [584 kB]
Fetched 584 kB in 3s (223 kB/s)
(Reading database... 139566 files and directories currently installed.)
Preparing to unpack .../linux-firmware-amd-misc_20260319.git217ca6e4-0ubuntu1.1_all.deb...
Unpacking linux-firmware-amd-misc (20260319.git217ca6e4-0ubuntu1.1) over (20260319.git217ca6e4-0ubuntu1)...
Setting up linux-firmware-amd-misc (20260319.git217ca6e4-0ubuntu1.1)...
Processing triggers for dracut (110-11)...
update-initramfs: Generating /boot/initrd.img-7.0.0-29-generic
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service

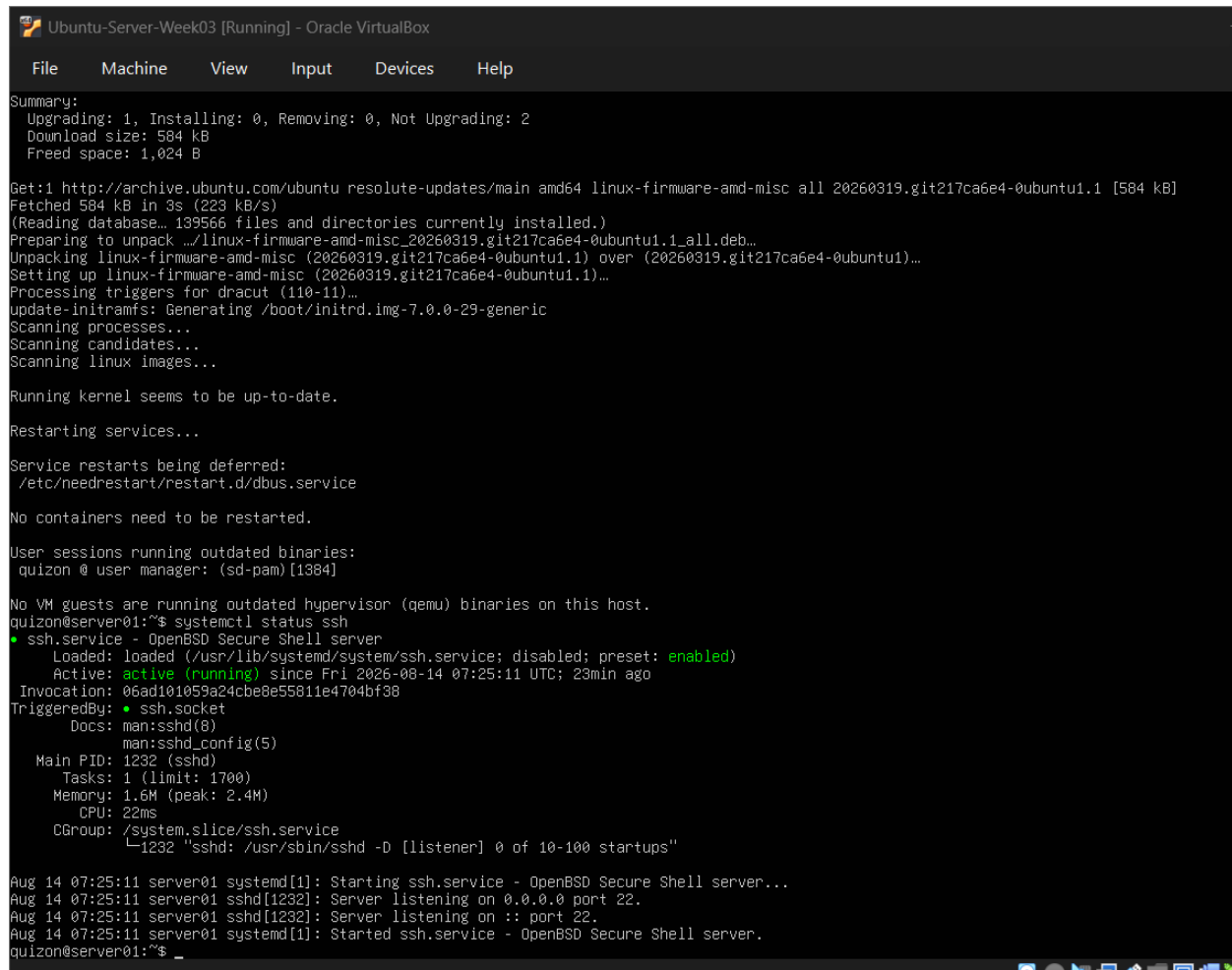
No containers need to be restarted.

User sessions running outdated binaries:
quizon @ user manager: (sd-pam)[1384]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
quizon@server01:~$
```

10. Verify the SSH Service

- To make sure everything matches. Verify the SSH service.
- Type “systemctl status ssh”
- It should say “active (running)”



```
Ubuntu-Server-Week03 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Summary:
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 2
  Download size: 584 kB
  Freed space: 1,024 B

Get:1 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-amd-misc all 20260319.git217ca6e4-0ubuntu1.1 [584 kB]
Fetched 584 kB in 3s (223 kB/s)
(Reading database... 139566 files and directories currently installed.)
Preparing to unpack .../linux-firmware-amd-misc_20260319.git217ca6e4-0ubuntu1.1_all.deb...
Unpacking linux-firmware-amd-misc (20260319.git217ca6e4-0ubuntu1.1) over (20260319.git217ca6e4-0ubuntu1.1)...
Setting up linux-firmware-amd-misc (20260319.git217ca6e4-0ubuntu1.1)...
Processing triggers for dracut (110-11)...
update-initramfs: Generating /boot/initrd.img-7.0.0-29-generic
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
  /etc/needrestart/restart.d/dbus.service

No containers need to be restarted.

User sessions running outdated binaries:
  quizon @ user manager: (sd-pam)[1384]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
quizon@server01:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-08-14 07:25:11 UTC; 23min ago
   Invocation: 06ad101059a24cbe8e55811e4704bf38
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 1232 (sshd)
     Tasks: 1 (limit: 1700)
    Memory: 1.6M (peak: 2.4M)
       CPU: 22ms
   CGroup: /system.slice/ssh.service
           └─1232 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 14 07:25:11 server01 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 14 07:25:11 server01 sshd[1232]: Server listening on 0.0.0.0 port 22.
Aug 14 07:25:11 server01 sshd[1232]: Server listening on :: port 22.
Aug 14 07:25:11 server01 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
quizon@server01:~$ _
```

Configuration Details

1. Profile and System Identity

- Host Name: server01 (sample)
- OS Edition: Ubuntu Server 25.04 LTS
- User Account: Quizon (sample)
- Network Interface: enp0s3

2. Virtual machine Hardware Configuration

- Hypervisor: Oracle VM VirtualBox
- RAM: 4GB / 4096 MB
- vCPU: 2 Processors
- Storage: 40 GB
- Network: NAT

Verification Commands

1. Login

- Log into the Ubuntu Server using the administrative user account created during setup.

2. Verify the Hostname

- Verify the system's hostname, make sure it matches the assigned computer identity.
- Type "hostname"

3. Verify the IP Address

- Verify the network interface and check the assigned IPv4 address.
- Type "ip addr"

4. Test the Internet Connectivity

- Confirm outbound internet access by sending ICMP packets to a public domain.
- Type "ping -c 4 google.com"

5. Update the Server

- Apply the latest system security patches and package updates.
- Type "sudo apt update" and "sudo apt upgrade -y"

6. Verify SSH Service

- Type "systemctl status ssh"

BIOS vs. UEFI Comparison

	BIOS	UEFI
Definition	BIOS (Basic Input/Output System) tells your computer's OS how to properly operate. It contains instructions on how to control various hardware components.	A modern and more flexible firmware interface that is designed to replace BIOS, acting as a lightweight mini-operating system between platform firmware and OS.
Boot Process	1. Power On – starts CPU executing BIOS 2. BIOS initializes hardware and runs POST 3. BIOS checks boot order and finds a bootable device 4. Loads MBR (Master Boot Record) from the device into memory 5. Transfers control to the MBR bootloader 6. Bootloader loads the OS	1. Power On – CPU starts UEFI firmware 2. Hardware initialization + drivers loaded 3. Boot Device Selection – reads BootOrder from NVRAM 4. Loads EFI bootloader from the EFI System Partition 5. Transfers control to the EFI application / OS bootloader 6. Bootloader loads the OS
Maximum Disk Support	2.2 Terabytes (TB) maximum addressable drive size using standard 512-byte sector sizing.	9.4 Zettabytes (ZB) (approx. 9.4 billion TB) maximum drive size.
Partition Style	MBR (Master Boot Record)	GPT (GUID Partition Table)
Security Features	Minimal firmware-level security; restricted primarily to basic supervisor/user power-on passwords.	Advanced hardware security including Secure Boot (cryptographic key validation)
Speed	Runs 16-bit	Runs 32-bit or 64-bit
Advantages	Widely compatible with older hardware and software	Faster boot, large disk support, better security, graphical UI +

		mouse, modular drivers, networking support in firmware, multi-boot friendliness.
Disadvantages	Smaller disk support, slow boot, more vulnerable to security threats.	Requires compatibility with the device, there are still flaws with the security.
Modern Usage	Mainly used only for old hardware	Standard on all new PCs

For many years, BIOS has been used as the fundamental firmware that provided computers' operating system with instructions to operate properly. However, as computers started to evolve more with modern technology, such as faster processors, larger storage, and growing security threats, the technology of the BIOS started to get left behind.

BIOS was created by Gary Kildall in 1975, but in 1981, IBM introduced BIOS as a proprietary technology. Years later, Phoenix teamed up with Compaq and Intel to develop the BIOS Boot Specification. As time goes by, its technology started to slow down. One of the biggest limitations was the storage. It was only able to handle 2 terabytes of storage, which in today's world, that is already too small for a lot of job industries. Also, manufacturers started making larger drives and SSDs that BIOS won't be able to manage properly.

As a solution to these problems, UEFI was introduced. It is the much modern version of the BIOS. It works extremely well with larger storage drives without having any problems. UEFI also starts computers much faster than what BIOS does in the past. Most importantly, UEFI includes a security feature called Secure Boot, this checks that the

software loading during the startup is trustworthy. This helps with protecting the computer against certain types of attacks that the old BIOS could not.

Boot Process Flowchart

Boot Process Flowchart

Ma. Lena Therese S. Quizon

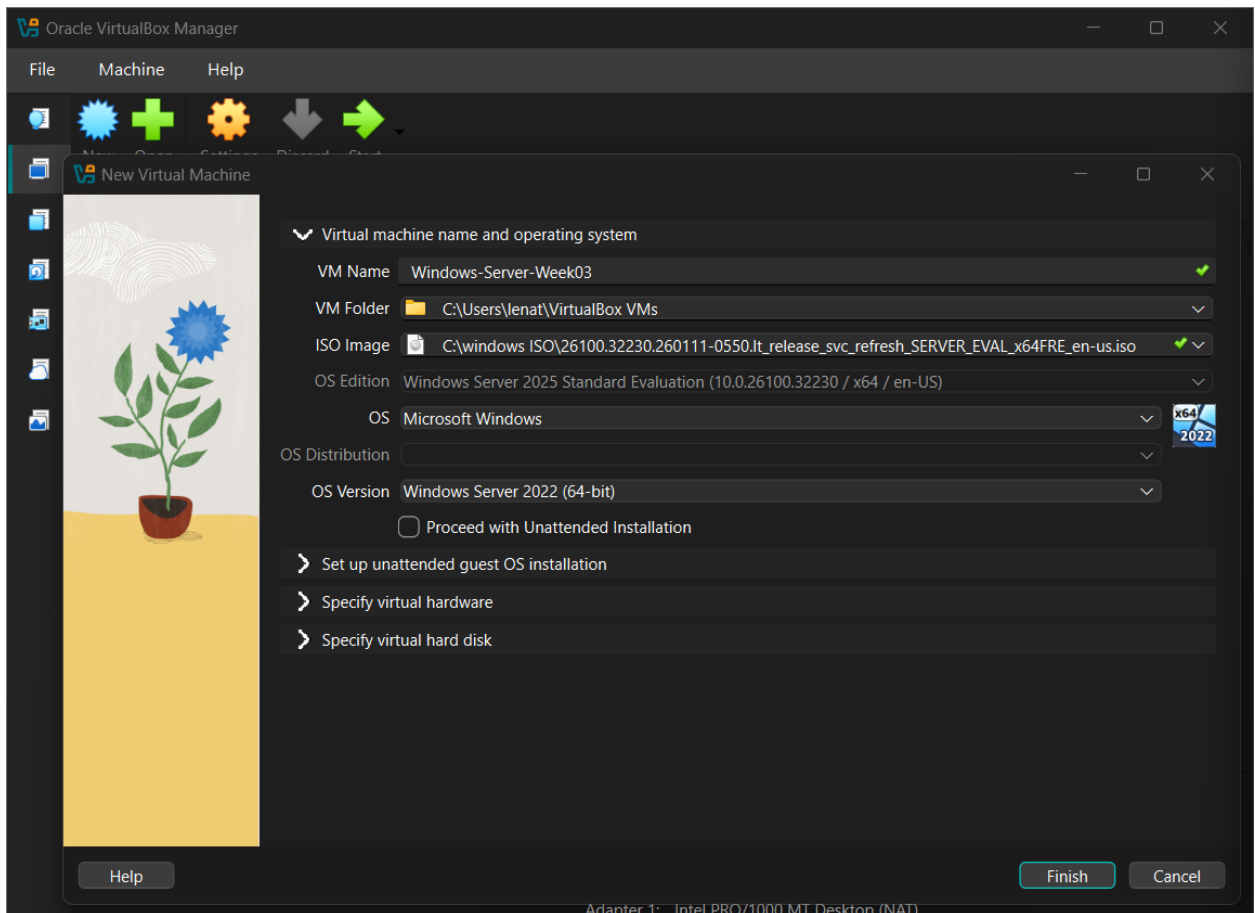


Windows Server Installation Summary

WINDOWS SERVER EVALUATION

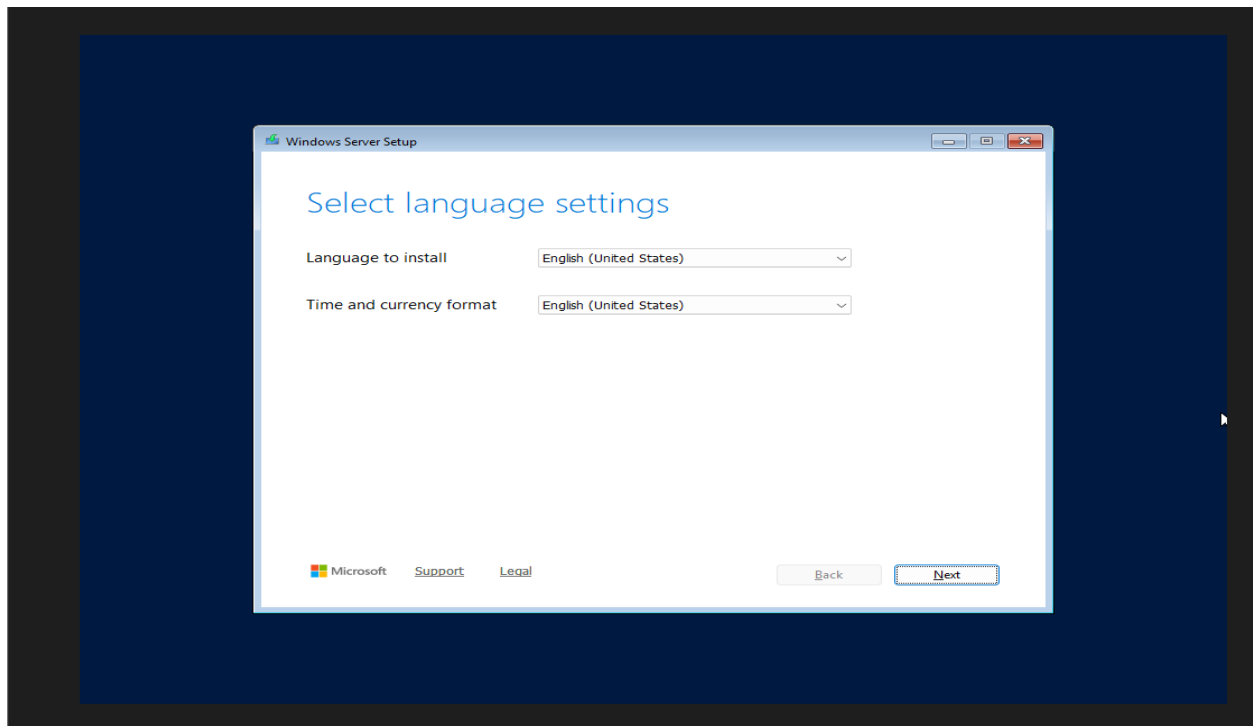
1. Create the Virtual Machine

- Open Oracle Virtual Machine.
- Press “New” to create a new virtual machine.
- Configure the VM name, select the ISO image, and uncheck the “Proceed with Unattended Installation” if it is checked.
- Click “Finish” once you are done.



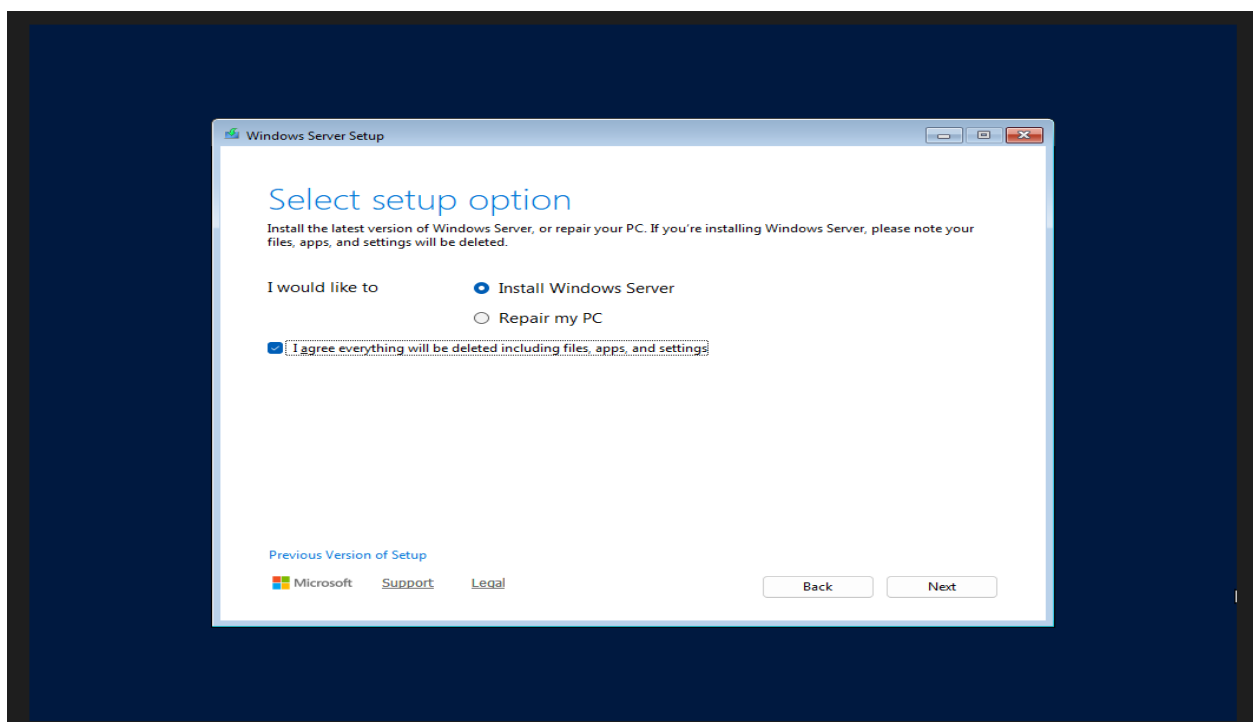
2. Start the Newly Created VM.

- Double Click the name of the VM to start.
- Once it is running. Select your preferred language and keyboard settings.



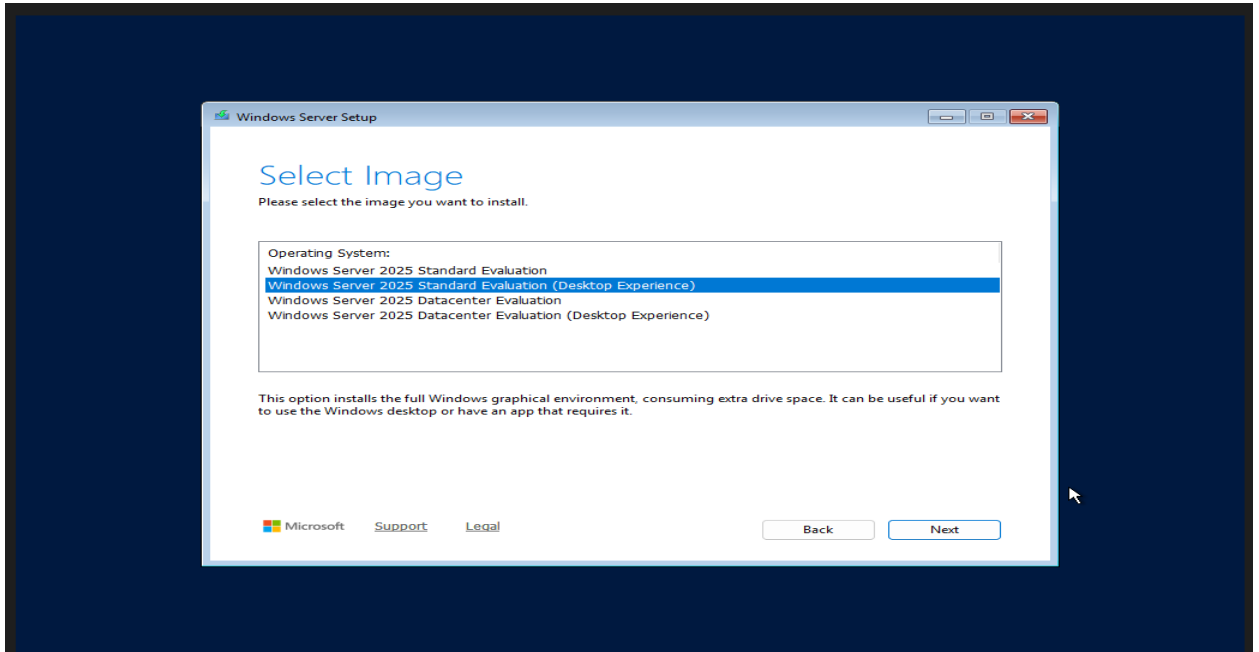
3. Select the Setup Option.

- Select the "Install Windows Server".
- Click the "I agree everything will be deleted including files, apps, and settings." so that you can click the "Next" button.



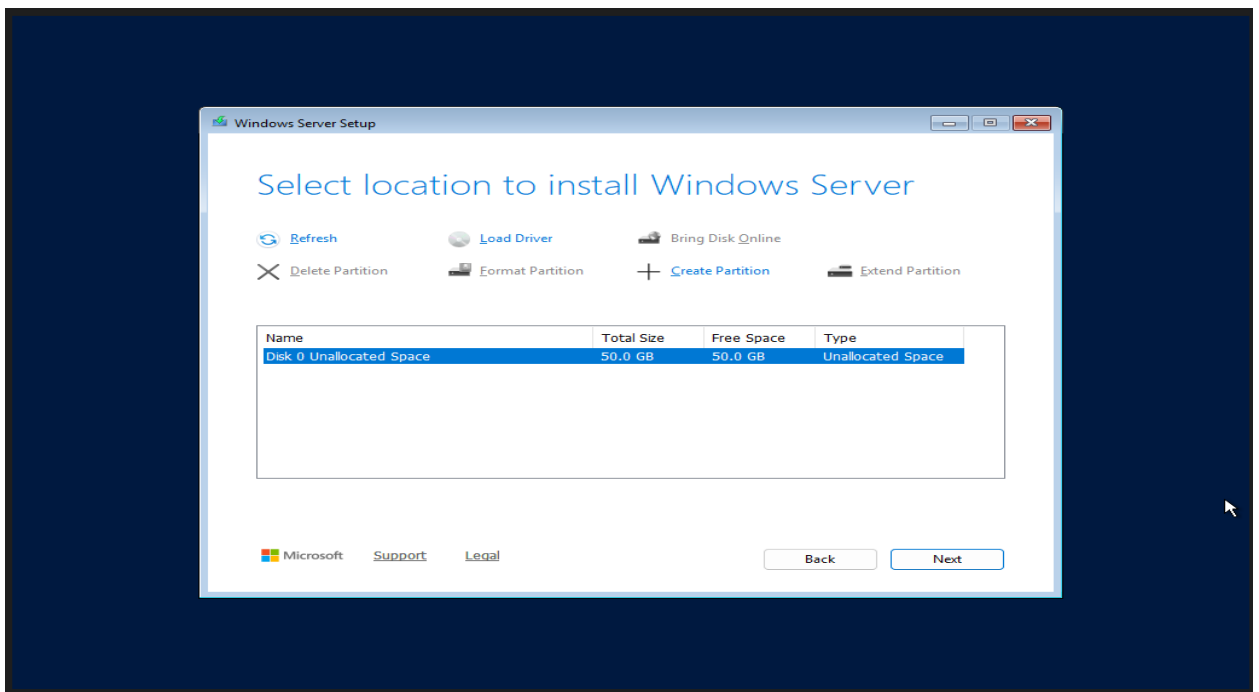
4. Select Image.

- Pick the “Windows Server 2025 Standard Evaluation (Desktop Experience).”
- After “Next”, accept the notice and license terms.



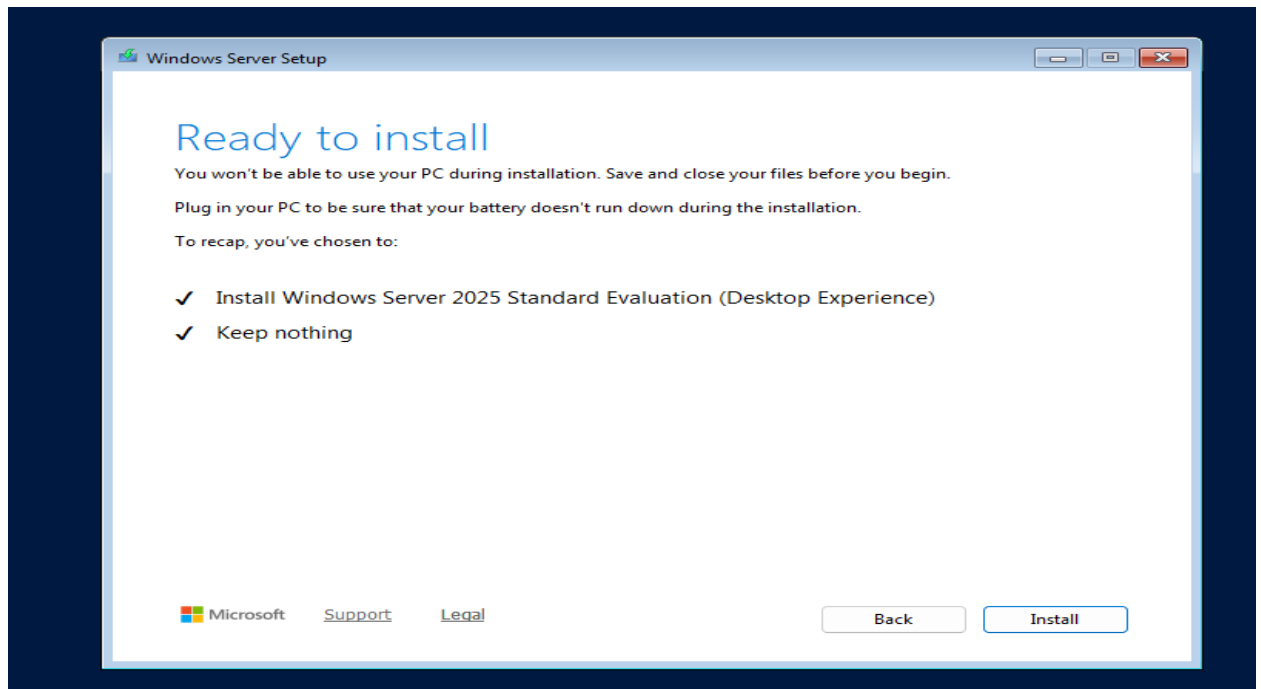
5. Select the Location to Install Windows Server.

- Click “Next” if the highlighted location is your preferred one.



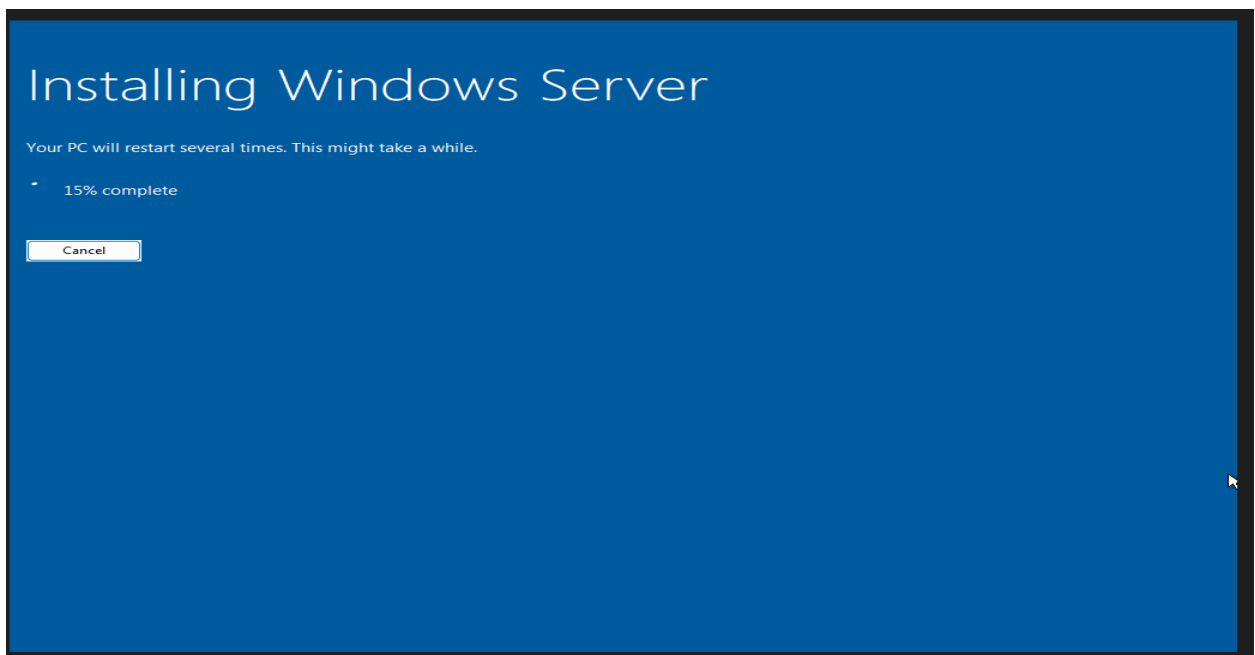
6. Ready to Install.

- Click “Install” once you’re ready.



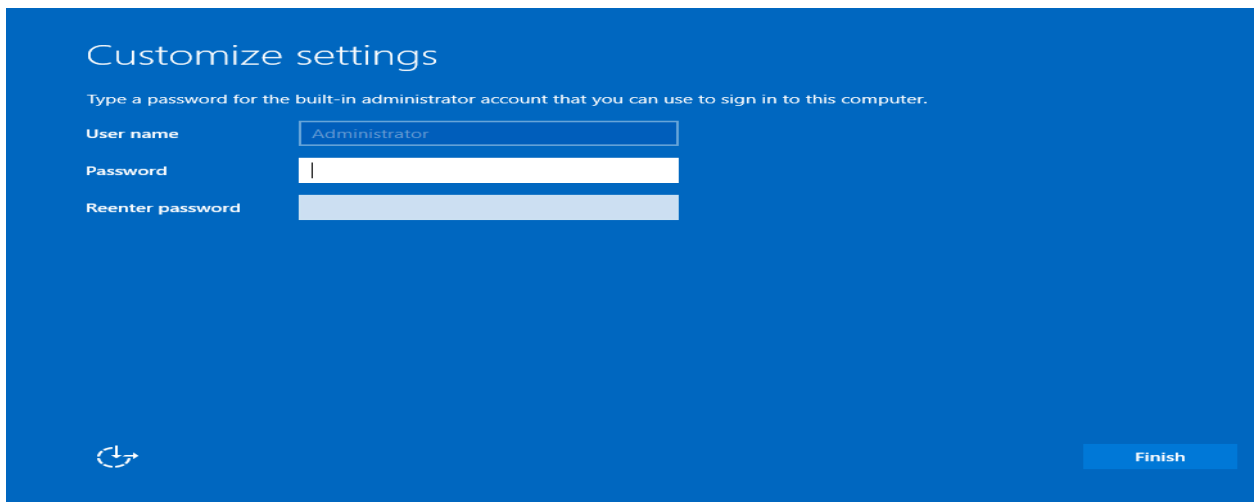
7. Wait for the Installation to Finish

- An installation window will pop up.
- Wait till the installation is completed.



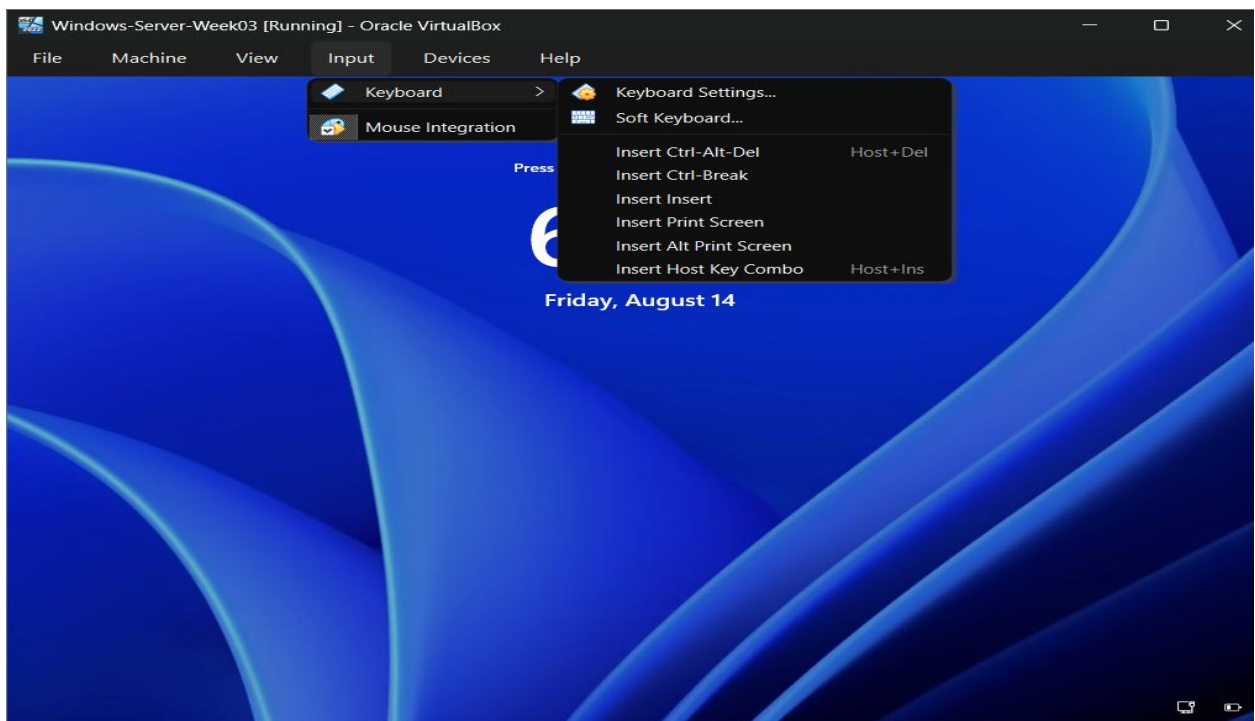
8. Customize Settings

- A window with a customize settings will pop up.
- Input the password. Make sure it has both letters and numbers.



9. Lockscreen Unlock

- Unlock the lockscreen by pressing the “Input” on the tab.
- Press “Keyboard”, and then select the “Insert Ctrl-Alt-Del”.



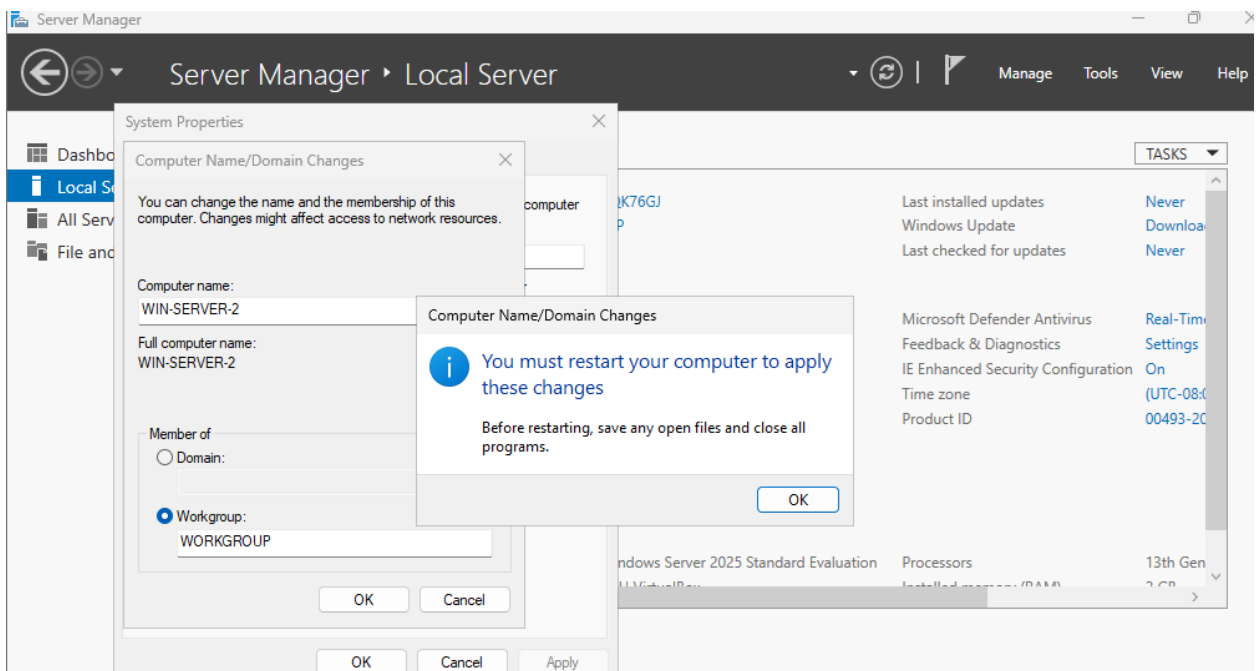
10. Login

- Login using the password you created.



11. Configure

- Server Manager will pop up.
- Go to "Local Server", and then click the "Computer Name"
- Click "Change" to rename the computer. Press "Ok" and a pop up will say that you must restart the computer. Press "Ok".
- Once the restart is done. You can proceed with other work.



OS Comparison Table

	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Proprietary/Commercial use	Free and open-source	Free and open-source
User Interface	Full graphical GUI	Command-line interface	Command-line interface
Package Management	Windows update, powershell, winget	APT (Advanced Package Tool) and snap packages	DNF/YUM
Security	Windows Defender, BitLocker, Active Directory	AppArmor, automatic security patches	SELinux (Mandatory Access Control)
Performance	Heavy resource usage	Lightweight, optimized for cloud and container workloads	Lightweight, optimized for enterprise workloads
Typical Enterprise Use Cases	Mixed environments, enterprise integration, commercial support, compliance scenarios, flexible licensing	Unified server operations, web dev, general servers, automated system hardening, and enterprise cloud maintenance	Enterprise servers, critical infrastructure, RHEL migration
Advantages	Familiar GUI, Microsoft ecosystem, great support for Windows applications.	Free, massive community, vast documentation, strong cloud support.	Binary compatibility with RHEL, consistent behavior, SELinux enforced by default

Disadvantages	High licensing cost, resource intensive, less stable than Linux	Steep learning curve, limited hardware support, software compatibility	Conservative software versions, smaller ecosystem, kernel patching requires reboots
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Determining the most suitable server operating system is a crucial decision that may affect the cost, security, performance, and long-term maintainability of an enterprise server system. Windows Server, Ubuntu Server, and the Rocky Linux can all serve enterprise-level performance, they still serve different organizational needs. Evaluating how these three platforms handle resources, software management, system security, and infrastructure can help organizations determine which operating system best aligns with their specific requirements, technical capabilities, and long-term infrastructure goals.

Windows Server is the most famous and common one you would see people use. It mainly stands out for its full graphical user interface and a deep integration with Microsoft ecosystem. While Ubuntu Server and Rocky Linux rely on lightweight command-line interfaces (CLI), Windows Server requires a significantly heavier hardware resource to power its background services. Window's security is handled through tools such as Windows Defender, BitLocker, and Active Directory, and the package management is handled by Windows Update, Powershell, and winget. Even though Windows is highly preferred by many due to its excellent support for Windows applications and mixed enterprise environments, it is more resource-heavy compare to Ubuntu Server and the Rocky Linux.

Ubuntu Server offers an open-source substitute which focuses on the use of modern cloud readiness. Identical to the Rocky Linux, both are open-source and uses a command-line interface. It manages software through the APT and Snap packages. Compared with the previous Window Server, Ubuntu is more lightweight and cost-effective, while offering newer software packages and stronger cloud optimization.

Lastly, similar to Ubuntu Server, Rocky Linux is also open-source and uses CLI. Even though Rocky Linux is lighter in resource than the Windows Server, and more

enterprise-focused than the Ubuntu Server, the software versions handled by Rocky Linux are more conservative and has a smaller ecosystem. It relies on the DNF/YUM package manager and enforces SELinux for strong mandatory access control.

In conclusion, the best choice comes down to what an enterprise actual needs. Many will choose Windows since it's ideal for Microsoft environments. Ubuntu is suited for organizations that prioritize cloud-optimized platforms, and flexibility. While Rocky Linux will be appealing for enterprises that needs a free, stable, RHEL-compatible system with robust security. Making the right choice will deliver the highest long-term value for an enterprise.

Problems Encountered

1. Large Download Size and Network Connectivity

- One of the major problems was the download size of the needed files. The Windows Server file was around 7.6 GB, considering how large the file was it took a while to download it.

2. APT Archive Download Failure

- When doing the 'sudo apt upgrade' command, it failed and showed a "Failed to fetch". The terminal also showed a 403 Forbidden error.

Solution Implemented

1. Large Download Size and Network Connectivity

- Managed to stable the connection by moving closer to our modem. I also had to restart the download multiple times when the download gets stuck in a specific percentage.

2. APT Archive Download Failure

- Re-synchronized the package index by running "sudo apt update" and executed "sudo apt upgrade --fix-missing" to force the package manager to bypass broken mirror links and complete the system update.

Lessons Learned

1. Be familiar with the interface of the command-line as it can be a bit confusing, compared to a full graphical interface like the Windows Server.
2. When an error happens while doing commands on the Ubuntu Server CLI, try to search on the web on how to solve that. Don't be afraid also to ask chatbots on how to fix those problems.
3. Remember to be patient when downloading large files. Large ISO files take time to download, especially when the internet connection is not stable. Constantly interrupting the download process will only lengthen the wait time.

References

BIOS:

- <https://www.lenovo.com/ph/en/glossary/what-is-a-bios/>

UEFI:

- <https://www.hp.com/hk-en/tech-takes/software/explainer/what-is-uefi.html>
- Max Disk Drive/Partition Style:
https://uefi.org/sites/default/files/resources/UEFI_Drive_Partition_Limits_Fact_Sheet.pdf
- <https://www.techtarget.com/whatis/definition/Unified-Extensible-Firmware-Interface-UEFI>

BIOS History:

- <https://zachary-brown.com/blog/bios/>
- <https://medium.com/@pratham.tomar23/understanding-bios-the-hidden-brain-that-boots-your-computer-969c35ce2707>

Windows Server:

- <https://learn.microsoft.com/en-us/windows-server/get-started/overview>

- <https://blog.webhostingworld.net/the-pros-and-cons-of-different-server-operating-systems/>

Ubuntu Server:

- <https://ubuntu.com/server>
- <https://dev.to/regoanac/pros-and-cons-of-ubuntu-exploring-advantages-and-disadvantages-of-the-linux-operating-system-2a4m>

Rocky Linux:

- <https://tuxcare.com/blog/rocky-linux-vs-ubuntu/>